

Assignment #3

MG-2009 Data Analysis for Business -II

Sections: BS(Fin-Tech) All Sections

Submission Date: 28/4/2026

Total Marks:130 [125 Assignment Marks +5 Presentation Marks]

Instructions:

- Assignment should be done on A4 size paper
 - There is no tolerance for plagiarism
 - Write down your class section on top right corner of the title page
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Q#1A manufacturing company produces the same product using two different machines and wants to compare the consistency of their output. Machine A produced a sample of 40 items with a variance in weight of 16 grams², while Machine B produced a sample of 35 items with a variance of 25 grams². Since lower variability indicates more consistent quality, management is interested in determining whether there is a significant difference in the variability of the two machines. To make this decision, the quality control team constructs a 95% confidence interval for the ratio of the two population variances and uses it to assess whether one machine is more consistent than the other. [10 marks]

Q#2: A company is evaluating the quality control of package weights produced by two different factories. Factory A has been in operation for many years, while Factory B uses more modern equipment. Management wants to determine whether there is a difference in the average weight of packages produced by the two factories. To investigate this, random samples of packages were selected from each factory, and their weights (in grams) were recorded. The sample data are as follows:

- Factory A: 118, 129, 132, 127, 113, 129, 112
- Factory B: 132, 121, 111, 123, 132, 121, 128, 113, 134, 132, 110, 108, 126

Based on this information, test the hypothesis at an appropriate level of significance to determine whether the mean weight of packages produced by Factory A is greater than that of Factory B. [15 Marks]

Q#3: The number of grams of carbohydrates contained in 1-ounce servings of randomly selected chocolate and nonchocolate candy is listed here. Is there sufficient evidence to conclude that there is a difference between the variation in carbohydrate content for chocolate and nonchocolate

candy? Use $\alpha=0.10$.

Chocolate 29 25 17 36 41 25 32 29 38 34 24 27 29

Nonchocolate 41 41 37 29 30 38 39 10 29 55 29

[10 marks]

Q#4 A financial technology company is analyzing the consistency of transaction processing times across two different digital payment systems. The company suspects that the newer system, System A, may have more variability in

processing times compared to the older, more stable System B. To investigate this, random samples of transaction times were collected from both systems. A sample of 30 transactions from System A shows a variance of 8,324 (milliseconds²), while a sample of 30 transactions from System B shows a variance of 2,862 (milliseconds²). At a significance level of $\alpha=0.05$, can the analyst conclude that the variation in transaction processing times for System A is greater than that for System B? Use an appropriate statistical test to justify your answer.

[10 marks]

Q#5: A fintech company is comparing the average transaction values processed through two different digital payment platforms, Platform A and Platform B. Due to differences in user behavior and system design, the variability in transaction amounts is believed to differ between the two platforms. To investigate this, independent random samples were collected. For Platform A, a sample of 20 transactions showed a mean value of \$520 with a standard deviation of \$80. For Platform B, a sample of 25 transactions showed a mean value of \$480 with a standard deviation of \$150. At a significance level of $\alpha=0.05$, test whether there is a significant difference in the mean transaction values between the two platforms.

[15 marks]

Q#6: A fintech company wants to examine whether the choice of payment method is independent of customer age group. A random sample of customers was classified according to their preferred payment method (Credit Card, Mobile Wallet, Bank Transfer) and age group (Under 30, 30–50, Above 50). The observed data are summarized below:

	Under 30	30–50	Above 50
Credit Card	40	35	25
Mobile Wallet	50	30	20
Bank Transfer	30	45	35

At the 0.10 significance level, test whether payment method preference is independent of age group.

[15 marks]

Q#7: A digital lending platform is analyzing whether loan approval decisions are independent of applicants' employment status. A sample of applicants is categorized as Employed, Self-Employed, or Unemployed, along with whether their loan was Approved or Rejected. The data are given below:

	Approved	Rejected
Employed	120	30
Self-Employed	80	40
Unemployed	50	60

Using a 5% level of significance, test whether loan approval is independent of employment status. [15marks]

Q#8: A fintech firm is studying whether fraudulent activity is related to the type of transaction. Transactions are categorized as Online Purchase, ATM Withdrawal, or Bank Transfer, and whether they were flagged as Fraudulent or Legitimate. The observed frequencies are:

	Fraudulent	Legitimate
Online Purchase	70	230
ATM Withdrawal	40	260
Bank Transfer	30	370

At the 0.001 significance level, test whether fraud occurrence is independent of transaction type. [15marks]

Q#9: A company has three manufacturing plants and wants to determine whether there is a significant difference between the average ages of workers at the three locations. The following data are the ages of five randomly selected workers at each plant. Perform a one-way ANOVA to determine whether there is a significant difference between the mean ages of the workers at the three plants. Use $\alpha = 0.01$ and note that the sample sizes are equal.

Worker ages		
Plant 1	Plant 2	Plant 3
29	32	25
27	33	24
30	31	24
27	34	25
28	30	26

[20 marks]